

Name: _____

Total: 18 points

Setup. As before, let $F : M \rightarrow M$ be a $\mathcal{C}^2$ map on a compact set $M \subset \mathbb{R}^d$. Fix an orbit $\{x_t\}_{t \geq 0}$ defined by $x_t := F(x_{t-1})$, and let $A_t := \nabla F(x_t)$ denote the $d \times d$ Jacobian matrix evaluated at x_t .

1. **(2 points)** A vector $v_t \in \mathbb{R}^d$ is called an *infinitesimal linear perturbation* at time t if the state x_t is perturbed to $x_t + \epsilon v_t$ for an arbitrarily small $\epsilon > 0$. Equivalently, v_t belongs to the *tangent space* $T_{x_t}M$ at x_t .

Starting from a given v_0 , derive the following recursive equation for the evolution of tangent vectors:

$$v_{t+1} = A_t v_t. \tag{1}$$

2. **(6 points)** Suppose we train a neural network F_{nn} by minimizing the mean squared error

$$R(F_{\text{nn}}) = \mathbb{E}_{x \sim \mu} \|F(x) - F_{\text{nn}}(x)\|^2,$$

where μ is an invariant measure of F . Let $\{y_t = F_{\text{nn}}^t(x_0)\}$ be the learned orbit, starting from the same initial condition x_0 as the true orbit.

For each of the following statements, determine whether it is true or false. Justify your answer.

- (a) If x_t converges to a fixed point, then $\|v_t\| \rightarrow 0$ for every choice of v_0 .

- (b) Let $w_{t+1} = B_t w_t$, where $B_t := \nabla F_{\text{nn}}(y_t)$. If both x_t and y_t converge to the same fixed point, then setting $w_0 = v_0$ implies $\|w_t\| \rightarrow 0$.

- (c) Suppose $\|v_t\| \sim \mathcal{O}(e^{\lambda t})$. If $\lambda < 0$, then for any $\delta > 0$, there exists an $\epsilon > 0$ such that $\|y_t - x_t\| < \delta$ for all t , whenever $R(F_{\text{nn}}) < \epsilon$.

3. (10 points) The figure below shows orbits of the Lorenz '63 system, a three-variable model introduced to demonstrate that the Earth's climate system can exhibit chaotic behavior.

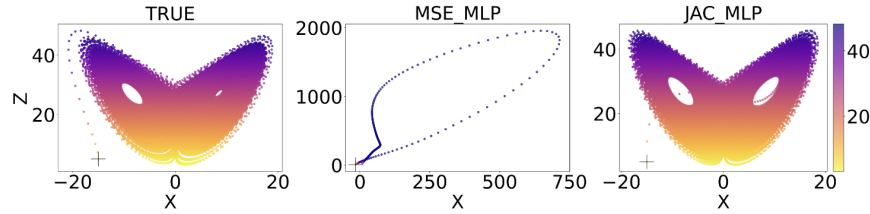


Figure 1: Left: true orbit of F . Center: orbit of F_{nn} , trained to minimize R . Right: orbit of G_{nn} , trained to minimize J .

The model F_{nn} minimizes the mean squared error R defined above. A second model, G_{nn} , minimizes the Jacobian-regularized objective

$$J(G) = R(G) + \alpha \mathbb{E}_{x \sim \mu} \|\nabla F(x) - \nabla G(x)\|^2,$$

where $\alpha > 0$ is a tuning parameter.

- (a) The true system F is chaotic: for Lebesgue-almost every v_0 , the tangent vector grows as $\|v_t\| \sim \mathcal{O}(e^{\lambda_1 t})$ with $\lambda_1 > 0$ for large t . Looking at the center figure, do you expect the tangent vector w_t along the orbit of F_{nn} to also exhibit exponential growth? Explain.
- (b) Will infinitesimal linear perturbations grow exponentially along the orbit of G_{nn} (right panel)? Explain.

- (c) Suppose G_{nn} preserves the measure μ . True or false: for any $x_0 \sim \mu$, although the orbits $\{G_{\text{nn}}^t(x_0)\}$ and $\{F^t(x_0)\}$ do not match pointwise, their empirical distributions converge to the same distribution as $t \rightarrow \infty$. Justify your answer.

- (d) Consider the following iterative procedure for computing the largest Lyapunov exponent:

Initialize. Choose $v_0 \neq 0 \in \mathbb{R}^d$ and set $\bar{\lambda}_1 = 0$. Repeat Steps 1–3 for $t = 1, 2, \dots, T$.

Step 1. Evolve the tangent vector: $v_t = A_t v_{t-1}$. Set $\eta_t = \|v_t\|$.

Step 2. Update the Lyapunov exponent estimate: $\bar{\lambda}_1 = (\bar{\lambda}_1 + \log \eta_t)/T$.

Step 3. Normalize: $v_t \leftarrow v_t/\eta_t$.

For large T , $\bar{\lambda}_1$ converges to the largest Lyapunov exponent—the asymptotic exponential growth rate of tangent vectors. In ergodic systems like the one considered here, this quantity is independent of the particular orbit $\{x_t\}$.

Now suppose $C(x) := \nabla G_{\text{nn}}(x)$ is a good approximation of $\nabla F(x)$ for all x . If you repeat the above procedure using F_{nn} in place of F , do you expect to recover the same largest Lyapunov exponent? What if you use G_{nn} instead? Justify your answers.